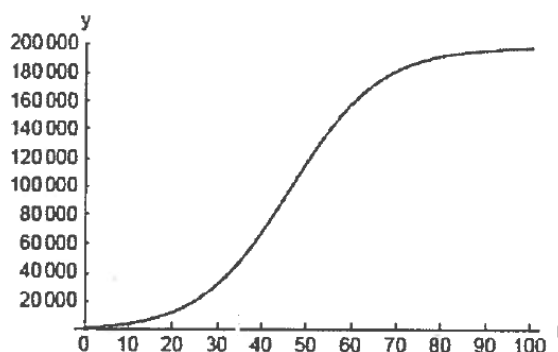


Problem 11

Problem 11

Problem 1 The total number of people, N who have contracted a common cold by a time t days after its outbreak on an island is given by $N = N(t) = \frac{200000}{1 + 100e^{-0.1t}}, t \geq 0$.

- (a) Evaluate and interpret the limit $\lim_{t \rightarrow \infty} N(t)$
- (b) How long will it take for the number of people who have contracted the cold to reach 40,000?
- (c) The graph of the function N on the interval $[0, 100]$ is given below. Sketch (as best you can) the graph of its derivative, $N'(t)$.



- (d) Calculate $N'(t)$. What does $N'(t)$ represent?
- (e) Evaluate and interpret the limit $\lim_{t \rightarrow \infty} N'(t)$
- (f) Find the average growth rate of the number of people who have contracted the disease during the time interval $[5, 6]$ (or during the sixth day after the outbreak).
- (g) Find the instantaneous growth rate of the number of people who have contracted the disease for $t = 5$ (round to a whole number).